TrES-3b

R-0386

Benchmark run · workflow W8-benchmark-deep · record `deep-bench_TRES-3_250829` · created 2026-07-18T21:56:30+00:00

BENCHMARK: NON-DETECTION

Results

Mid-Transit Time (Tmid)	2460916.686 +/- 0.0019 BJD_TDB
Ratio of Planet to Stellar Radius (Rp/R*)	0.192 +/- 0.058
Transit depth (Rp/Rs)^2	3.67 +/- 2.23 [%]
Orbital Inclination (inc)	84.27 +/- 2.8
Airmass coefficient 1 (a1)	1.41 +/- 0.21
Airmass coefficient 2 (a2)	-0.26 +/- 0.13
Scatter in the residuals of the lightcurve fit is	15.63 %
Best Comparison Star	#1 - [436, 415]
Optimal Aperture	1.61
Optimal Annulus	15.05
Transit Duration (day)	0.069 +/- 0.018

Accuracy vs published ephemeris (ExoClock/NEA)

Rp/R■ fit vs published	0.192 ± 0.058 vs 0.16309 (deviation 0.37σ)
Duration fit vs published	0.069 d vs 0.0567 d (deviation 0.51σ)
Mid-time O–C	-5.3 min
Depth SNR	2.5
Residual scatter	15.63 %

Light curve

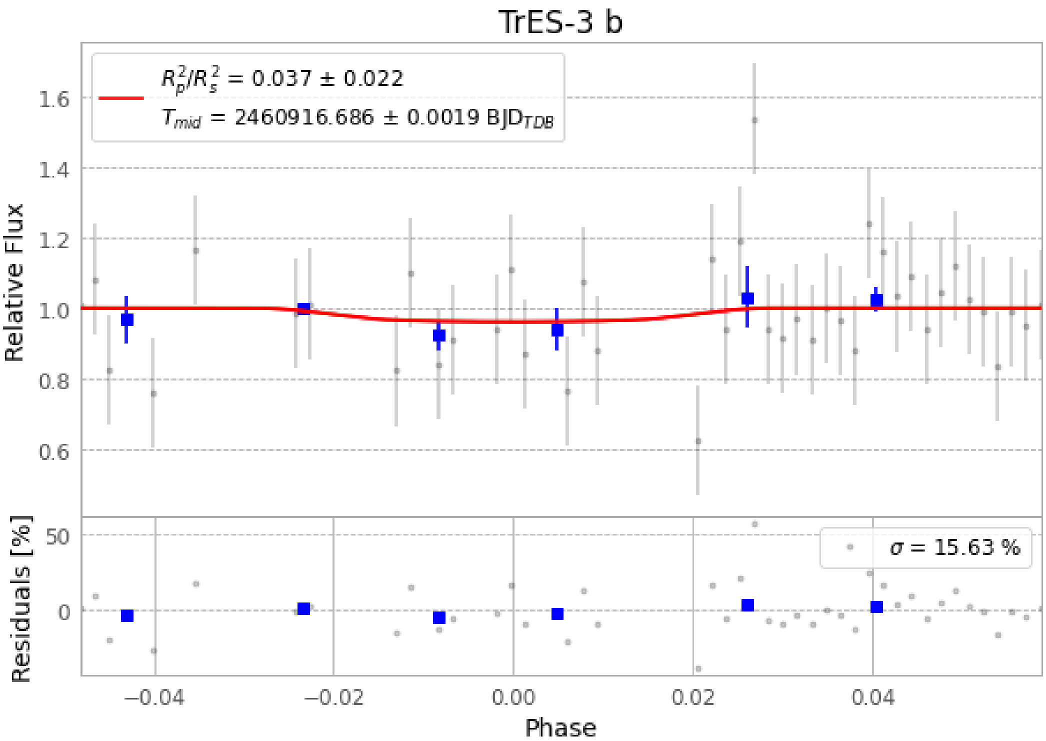

Figure 1 — FinalLightCurve_TrES-3 b_2025-08-28.png (SHA-256 94cfbda8c5f70612…)

Method & software

EXOTIC 4.3.1 aperture photometry with nested-sampling transit fit (ultranest 3.6.5); priors from the NASA Exoplanet Archive. Camera CCD, binning 1x1, filter CV, target pixel [328, 257], 3 comparison star(s). Exact configuration: `parameters.inits_json` in record.json (SHA-256 d9e04d1f333e1c58…).

`exotic 4.3.1 · astropy 6.1.7 · photutils 2.0.2 · numpy 1.26.4 · scipy 1.14.1 · twirl 0.5.1 · astroquery 0.4.11 · ultranest 3.6.5 · python 3.12.13 · pipeline_commit 8350c6b`

Data provenance

68 FITS frames (45 MB), `TRES-3250829025415.FITS` → `TRES-3250829061515.FITS`. Every input frame is SHA-256-fingerprinted in record.json; raw frames available on request and verifiable against that manifest. Artifacts: bench-TRES-3-250829.log (f8968d775627…), FinalLightCurve_TrES-3 b_2025-08-28.png (94cfbda8c5f7…), FinalLightCurve_TrES-3 b_2025-08-28.csv (ce795776ac77…)

Compute resources

Wall time 200.4 s · peak memory None MB · host mac-mini-m1-8gb (macOS-26.4-arm64-arm-64bit)